

real returns on stocks, establishing that the CAPE ratio was a significant variable predicting long-run stock returns.^{[9](#)}

Implied Cost of Capital (ICC)

- Li, Ng, and Swaminathan (2013): *“Theoretically, the implied cost of capital (ICC) is a good proxy for time-varying expected returns. We find that aggregate ICC strongly predicts future excess market returns at horizons ranging from one month to four years... We also find that ICCs of size and B/M portfolios predict corresponding portfolio returns.”*^{[10](#)}

Sum of the Parts Forecasting

- Ferreira and Santa-Clara (2011): *“We propose forecasting separately the three components of stock market returns—the dividend–price ratio, earnings growth, and price–earnings ratio growth—the sum-of-the-parts (SOP) method. Our method exploits the different time-series persistence of the components and obtains out-of-sample R-squares (compared with the historical mean) of more than 1.3% with monthly data and 13.4% with yearly data.”*^{[11](#)}

Technical Indicators

Technical indicators have been written off as heresy by many academics. The biggest complaint is that technical trading rules rely on past prices to predict futures returns. This concept flies in the face of the efficient market hypothesis, and therefore requires researchers to jump over a high bar to get their research published in a peer-reviewed journal. As academics, we understand this skepticism with technical trading rules, but as empiricists, we are willing to give any idea a fair shot. Below are some of the more interesting research papers on the subject.

Time Series Momentum

- Moskowitz, Ooi, and Pedersen (2010): *“We document significant ‘time series momentum’ in equity index, currency, commodity, and bond futures for each of the 58 liquid instruments we consider... A diversified portfolio of time series momentum strategies across all asset classes*